# Chapter 16 worked solutions • Virtual machines

## OS16-E01 • Modes and virtual interrupts

1. No. The VMM emulates the guest kernel's legal CLI by changing this guest's virtual interrupt-enable state and controlling its maskable interrupt injection; the host must still serve other guests and devices. In the stated protected-mode configuration, EFLAGS.VM=0, CR4.PVI=0, and application CPL=3 exceeds IOPL=0, so CLI raises #GP(0). If intercepted, the VMM injects that guest exception as specified, rather than granting the operation. Other settings, including PVI, may produce different behavior.
2. Without a trap the VMM cannot update the virtual state, and the guest may observe incorrect semantics. Binary translation, cooperative hypercalls, or hardware interception can mediate it; no trap does not mean no emulation is needed.

**Check.** Guest ring 0 and VMM root context are different axes.

## OS16-E02 • Addresses and permission intersection

1. Displacement0xABC gives GPA0x7ABC, HPA0x12ABC. Read passes both; EPT denies write despite guest permission.
2. Because the new GFN does not map to MFN 0x12, stale shadow translation can reach the old host page. Track guest updates, rebuild the shadow composition, and invalidate old translations under architecture rules. Hardware two-stage also needs its own cache/invalidation rules. GVA, GPA, HPA are distinct.

## OS16-E03 • General nested-walk cost

1. Each guest PTE needs 4 EPT reads plus one guest PTE read; three cost 15. The final GPA needs 4 more EPT reads: 19 table reads plus one data read gives 20. The general table-read count is $(m+1)(n+1)-1$. Possible accessed/dirty writes are excluded.
2. Every guest PTE itself resides in guest physical memory and needs second-stage translation. Cache hits can reduce actual reads; 19 applies only to this cold-cache model. Separate table and data references.

## OS16-E04 • Four-slot PV ring

1. Queued7−6=1, outstanding6−5=1, response5−4=1, used7−4=3, free1. Next slot7 mod4=3. Use monotonic counts, not differences of modulo pointers.
2. Submission makes req_prod8, used4, full. Consuming response advances rsp_cons4→5, used3, free1, allowing one more. Advancing req_cons alone releases no frontend capacity.
3. Fill descriptor then release-publish req_prod; backend acquire-observes before reading. Fill response before publishing rsp_prod; frontend consumes then advances rsp_cons. Notifications replace neither predicates nor ordering. Under one lock, recheck the predicate, enqueue the waiter, and release the lock atomically with sleeping; the producer updates the predicate and signals under the same synchronization. A final check followed by a separate sleep still permits a lost wakeup. Buffers live through completion. Out-of-order completion needs another model.

## OS16-E05 • Finite two-layer scheduling trace

1. A runs0–10 and20–30, finishing30. B runs10–20 and30–40, finishing40. This finite trace ends when both jobs finish, so it alone does not establish a long-run share. If both remain runnable indefinitely and alternate forever, each gets 50%; phase still affects finite completion times.
2. No. Guest schedules within vCPU allocations; it cannot force host pCPU service. Stronger latency requires host real-time/isolation/admission guarantees. CPU runtime and wall response differ.

## OS16-E06 • Balloon conservation

1. Demand8192 exceeds guest-available7168 by1024. Ballooning512 each gives targets3584+3584=7168, plus VMM 1024=8192. This conserves targets, not a claim that original guest 8192 plus VMM 1024 already fit. Inflation reduces backing demand without creating RAM.
2. No. Driver reserves guest pages and reports them; VMM reclaims backing. The accidental-access response is implementation-specific; in a fault-and-reback model the VMM must supply safe backing before access, never another guest's frame. Cache reclamation/pageout costs remain. Guest reservation does not mean host backing is unreclaimable.

## OS16-E07 • Sharing and COW

1. A hash only selects candidates. Lock or write-protect both pages and related mappings, complete required CPU TLB/EPT invalidations, and stop or isolate DMA and other writers. Compare all bytes while this protection holds, then establish both mappings read-only to h9. Free h11 only after all other references, pins, and in-flight accesses have ended. On B's later write fault, allocate h12, copy X, update B's mapping and required translations, synchronize, then permit the write Y; A still sees X.
2. B's pre-merge writable entry still points to h11: it can modify the old page or a reused frame, not directly h9. An old writable A translation to h9 can instead modify the shared page. Complete applicable TLB and second-stage invalidation and synchronization before stable comparison and sharing.

## OS16-E08 • Migration convergence and pause

1. The stated model dirties exactly 32 MiB/s of distinct data per round and stops all writers at freeze. R0=512 copies4 s→R1=128; next1 s→R2=32; next0.25 s→R3=8. Freeze copies8 in 0.0625 s. Total680 MiB,5.3125 s, pause0.0625 s.
2. Ratio=d/B=1, so512 does not shrink. Bound rounds, reduce dirtying, extend pause, or change strategy. Short downtime is not universal.
3. Two active copies with one identity can create duplicate effects, or split brain. Explicit ownership/commit/recovery is required. A normal trace does not prove partition safety. Compatible CPU/devices and shared storage are this model's premises.

## OS16-E09 • Snapshots and external effects

1. Active read 0=A, read 1=C; base restores read 1=B. Mail remains sent, and rerunning can send twice. A snapshot is not an external-world rollback.
2. No. Files/chains/RAM/disk can represent different times. Shutdown or supported consistent snapshot plus application quiescence and all required metadata/chains is needed. Same-failure-domain snapshots are not independent backups. Crash and application consistency differ.

## OS16-E10 • Abstractions and bytecode

1. Another kernel requires system VM/emulator; cross-ISA binary needs emulation/translation; class requires JVM/runtime; same-kernel separation can use containment. Containers share kernel, system VMs have guest kernels, and JVM is not another general OS.
2. With at least two local-variable slots and the rest of the method valid: []→[2]→[2,3]→[5]→[], local1=5. Iadd needs two integers; empty stack violates verifier premises rather than authorizing arbitrary memory access. Type checking does not prove business authorization. The interface/isolation layers determine guarantees.

### References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 16. Xen SOSP 2003. ESX memory management OSDI 2002. JVM SE7 verifier.

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https://github.com/xinhi6716/computer-science-self-study

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